Εφαρμογή 1

$x_1 = (1, 2, 3)$
$x_2 = (2, 4, 6)$
$x_3 = (5, 7, 9)$
$x_4 = (0, 1, 1)$

a) $x_{12} = \sqrt{dx^2 + dy^2 + dz^2} =$
$= \sqrt{(2-1)^2 + (4-2)^2 + (6-3)^2} =$
$= \sqrt{1 + 4 + 9} = \sqrt{14} =$

$x_{13} = \sqrt{dx^2 + dy^2 + dz^2} =$
$= \sqrt{(5-1)^2 + (7-2)^2 + (9-3)^2} =$
$= \sqrt{16 + 25 + 36} = \sqrt{77}$

$x_{14} = \sqrt{dx^2 + dy^2 + dz^2} =$
$= \sqrt{(0-1)^2 + (1-2)^2 + (1-3)^2} =$
$= \sqrt{1 + 1 + 4} = \sqrt{6}$

$x_{23} = \sqrt{dx^2 + dy^2 + dz^2} =$
$= \sqrt{(5-2)^2 + (7-4)^2 + (9-6)^2} =$
$= \sqrt{9 + 9 + 9} = \sqrt{27}$

$x_{24} = \sqrt{dx^2 + dy^2 + dz^2} =$
$= \sqrt{(0-2)^2 + (1-4)^2 + (1-6)^2} =$
$= \sqrt{4 + 9 + 25} = \sqrt{38}$

$x_{34} = \sqrt{dx^2 + dy^2 + dz^2} =$
$= \sqrt{(0-5)^2 + (1-7)^2 + (1-9)^2} =$
$= \sqrt{25 + 36 + 64} = \sqrt{125}$

6) $X_{10} = \sqrt{dx^2+dy^2+dz^2} =$

$= \sqrt{(0-1)^2+(0-2)^2+(0-3)^2} =$

$= \sqrt{1+4+9} = \sqrt{14}$

$X_{20} = \sqrt{dx^2+dy^2+dz^2} =$

$= \sqrt{(0-2)^2+(0-4)^2+(0-6)^2} =$

$= \sqrt{4+16+36} = \sqrt{54}$

$X_{30} = \sqrt{dx^2+dy^2+dz^2} =$

$= \sqrt{(0-5)^2+(0-7)^2+(0-9)^2} =$

$= \sqrt{25+49+81} = \sqrt{155}$

$X_{40} = \sqrt{dx^2+dy^2+dz^2} =$

$= \sqrt{(0-0)^2+(0-1)^2+(0-1)^2} = \sqrt{2}$

7) $x = \frac{fX}{Z}$ $\quad y = \frac{fY}{Z}$ $\quad f = 1$

$X_1 = \frac{1}{3}$ $\quad y_1 = \frac{2}{3}$

$X_1' = \left(\frac{1}{3}, \frac{2}{3}\right)$

$X_2 = \frac{2}{6} = \frac{1}{3}$ $\quad y_2 = \frac{4}{6} = \frac{2}{3}$

$X_2' = \left(\frac{1}{3}, \frac{2}{3}\right)$

$x_3 = \frac{5}{9}$ $\qquad$ $y_3 = \frac{7}{9}$

$x_3' = \left(\frac{5}{9}, \frac{7}{9}\right)$

$x_4 = \frac{0}{1} = 0$ $\qquad$ $y_4 = \frac{1}{1} = 1$

$x_4' = (0,1)$

Παράδειγμα 2

$2 \cdot x_1 = (2,4,6)$
$2 \cdot x_2 = (4,8,12)$
$2 \cdot x_3 = (10,14,18)$
$2 \cdot x_4 = (0,2,2)$

Κάθε x αντιστοιχεί σε μία ευθεία, άρα με το να πολλαπλασιάσουμε τις ήδη υπάρχουσες συντεταγμένες των σημείων βρίσκουμε σημεία σε άλλο σημείο της ίδιας ευθείας.

$x_1 = \frac{1 \cdot 2}{6} = \frac{1}{3}$ $\qquad$ $y_1 = \frac{1 \cdot 4}{6} = \frac{2}{3}$

$x_2 = \frac{1 \cdot 4}{12} = \frac{1}{3}$ $\qquad$ $y_2 = \frac{1 \cdot 8}{12} = \frac{2}{3}$

$x_3 = \frac{1 \cdot 10}{18} = \frac{5}{9}$ $\qquad$ $y_3 = \frac{1 \cdot 14}{18} = \frac{7}{9}$

$x_4 = \frac{1 \cdot 0}{2} = 0$ $\qquad$ $y_4 = \frac{1 \cdot 2}{2} = 1$